

Spawtify: How to Build a Digital Platform for Animal Spotting

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Abstract—This report presents the design, implementation, and evaluation of Spawtify, a web-based platform that enables urban communities to record, share, and discover animal sightings within their neighborhoods. Motivated by New Zealand’s distinctive urban ecology, Spawtify addresses three critical community needs: facilitating lost pet recovery, fostering neighborhood engagement through wildlife observation, and supporting urban ecological awareness. The platform employs a modern technology stack comprising React.js for the frontend interface, Node.js with Express.js for backend services, MongoDB for flexible data storage, AWS S3 for scalable image management, and Leaflet for interactive mapping capabilities. Key technical contributions include dynamic geographic boundary definition through map-based user interaction, real-time geocoding for address-to-coordinate conversion, and secure cloud-based image storage integration. The development process utilized agile methodologies with continuous integration practices, comprehensive testing protocols, and AI-assisted development tools to enhance productivity while maintaining code quality.

I. INTRODUCTION

Picture this: you’re standing in your garden on a cold Auckland morning, clutching a cup of coffee that’s gone lukewarm while you were distracted by something extraordinary. A massive ginger cat—and I mean properly massive, like a small orange tiger—glides through your garden with the sort of effortless grace that would make a Jaguar E-type driver weep with envy. This isn’t just any garden prowler; this is a feline that moves like it owns the place, which, let’s be honest, it probably does.

Now, if you’re like me and grew up somewhere where gardens are about as common as hen’s teeth and pets are kept locked up tighter than the China Internet Firewall, this sight is nothing short of revolutionary. The cat leaps onto the fence with the casual elegance of a gazelle, pauses for a moment to survey its domain, and then vanishes like morning mist. Gone. Just like that. Leaving you standing there with your lukewarm coffee, wondering what the devil just happened and whether you’ve just witnessed something magical or simply Tuesday morning in New Zealand.

This is the moment when Spawtify was born. The genius of this moment wasn’t just in the observation; it was in the realization that nearly forty percent of New Zealand households own at least one cat (Companion Animals in New Zealand, 2020), and many of these feline overlords spend more time outdoors than some people spend at their actual jobs. Think about it: we live in an age where we can track our pizza delivery in real-time, monitor our heart rate while sleeping, and know exactly how many steps we’ve

taken to the refrigerator, yet we have absolutely no idea what our neighborhood cats are up to. Here we have an entire ecosystem of urban wildlife—cats, dogs, possums, birds, and the occasional confused duck family—wandering around our neighborhoods like they own the place, and we’re all just going about our business pretending we don’t care. But we do care, don’t we? We absolutely care.

The turning point came later that year when my garden became something of a wildlife theater. First came the duck family—ten ducklings following their mother in a perfect line, like a feathered convoy of adorable chaos. Then came Grace, a grey shorthair cat with the sort of regal bearing that suggested she was slumming it in our humble neighborhood. Grace became a regular visitor, turning up at predictable times like she was punching a time clock. We grew fond of Grace, as you do with any creature that brightens your day with its presence.

The plot thickened when we discovered Grace’s real identity. Her actual name was Jessie, and she belonged to a lovely couple who’d recently moved to the neighborhood. They were delighted that Jessie had made friends and was exploring her new territory, but they also harbored the sort of concerns that keep pet owners awake at night: What if she gets lost? What if she can’t find her way home? What if someone thinks she’s a stray and takes her in? These are the questions that haunt every pet owner who’s ever watched their beloved companion disappear over a fence.

This conversation sparked something bigger than just idle curiosity about neighborhood cats. It revealed a fundamental gap in how we interact with our urban environment. We have social networks for humans, tracking systems for packages, and apps for everything from ordering food to finding romance, but nothing that connects us to the animals that share our neighborhoods. Nothing that helps us understand the secret lives of urban wildlife. Nothing that bridges the gap between pet owners’ worries and neighbors’ willingness to help.

II. GOALS

Spawtify emerged from this realization with three core objectives that go far beyond simple pet tracking. First, we wanted to support New Zealand’s unique urban ecology by creating a platform where communities could collectively track birds, monitor stray animals, and document the presence of pests. This isn’t just about satisfying curiosity; it’s about understanding the complex web of urban wildlife that most of us never really see.

Second, we recognized the need for a genuinely effective system for reporting and finding missing pets. Traditional methods—those heartbreaking “LOST CAT” posters taped to lamp posts, frantic Facebook posts that get buried in the timeline, desperate calls to local shelters—are fragmented, inefficient, and often ineffective. We wanted to create something unified, immediate, and map-based that could harness the collective eyes and ears of an entire community.

Third, and perhaps most importantly, we wanted to encourage neighborhood engagement through a shared interest in local animals. There’s something uniquely powerful about animals as a medium for community connection. They transcend the usual barriers of age, background, and lifestyle. The businessman rushing to catch the morning train and the retired teacher tending her garden can both share a moment of wonder at spotting a tui in the oak tree, or concern about the skinny tabby that’s been hanging around the bus stop.

What makes Spawtify different from existing animal sighting platforms is its deliberate inclusivity and community focus. Take eBird (Sullivan et al., 2014), for instance—a brilliant platform that’s become the gold standard for bird observation data. eBird has revolutionized ornithology by crowdsourcing bird sightings from around the world, creating an incredibly rich dataset that helps scientists understand migration patterns, population changes, and habitat preferences. But eBird is fundamentally designed for bird enthusiasts, people who know their fantails from their silvereyes and can identify a call from fifty meters away.

Then there’s Dorsal (Araujo et al., 2020), which focuses on shark sightings to enhance beach safety. Dorsal serves a crucial function by providing real-time information about potentially dangerous marine life, helping beachgoers make informed decisions about when and where to swim. Both platforms excel at what they do, but they’re specialized tools for specific audiences with particular expertise.

Spawtify takes a different approach entirely. We’re not asking people to become expert naturalists or to learn the difference between a morepork and a ruru (though they’re actually the same bird, which is the sort of thing that makes you appreciate why not everyone wants to be a bird expert). We’re simply asking people to notice the animals around them and share what they see. The cat sunning itself on the neighbor’s fence. The hedgehog shuffling across the lawn at dusk. The family of ducks that inexplicably decided to set up shop in someone’s backyard stream.

III. IMPLEMENTATION HIGHLIGHTS

- Frontend: Built with React.js, Leaflet for interactive maps, and Tailwind CSS for styling.
- Backend: Node.js with Express.js, providing a RESTful API for data access and user authentication.
- Database: MongoDB, chosen for its flexibility and scalability.
- Image Storage: AWS S3, used for storing user-uploaded photos.

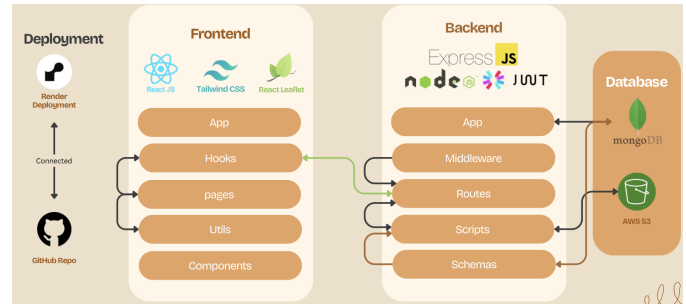


Fig. 1. Tech Stack Overview

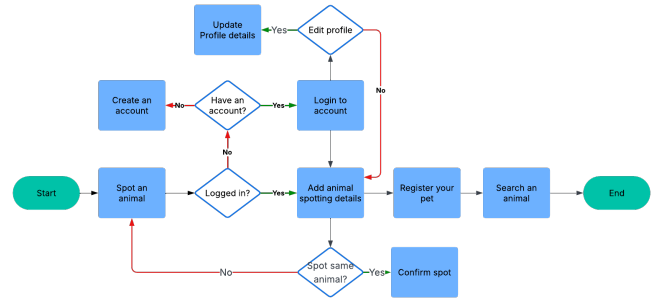


Fig. 2. The logic flow of the frontend

- Map Integration: Leaflet, an open-source JavaScript library for interactive maps.
- Deployment: On Render, the web app is automatically deployed every time a change is merged to our GitHub repository main branch. Check it out: <https://spawtify.onrender.com/>

Creating Spawtify required making some crucial architectural decisions that would determine whether our platform would be a sleek, efficient machine or a digital equivalent of a three-wheeled shopping trolley. We chose to build on a modern web stack that separates the frontend and backend cleanly, like a well-organized garage where you can actually find the tools you need. The frontend runs on React, which handles the user interface with the sort of smooth efficiency that makes you forget you’re interacting with a computer. We integrated React-Leaflet for the interactive maps because, let’s face it, everything is better with maps. Maps make data visual, immediate, and geographically relevant in a way that spreadsheets never could. We integrated Leaflet as our mapping solution because it’s open-source, flexible, and doesn’t require selling your soul to a major technology corporation like Google to use effectively. We styled frontend with Tailwind CSS, which I had to learn from scratch—a process that was initially about as enjoyable as learning to drive in a car with no power steering, but which ultimately provided the sort of precise control that makes the effort worthwhile.

The backend is powered by Node.js with Express.js, providing a RESTful API that handles data access and user authentication with reliability and scalability. We chose MongoDB for our database because its flexible schema structure

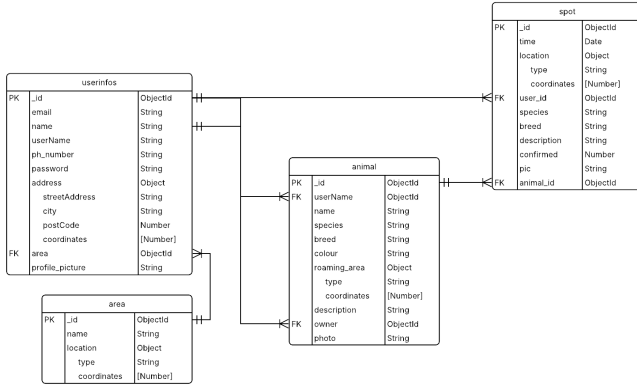


Fig. 3. Database Schema Model

accommodates the diverse types of data we’re collecting without forcing everything into relational tables and queries that take way longer to declare.

For image storage, we use AWS S3, which handles user-uploaded photos with the sort of industrial-strength reliability that you need when people are entrusting you with pictures of their beloved pets. User authentication is handled via JWT tokens, with middleware protecting sensitive routes. Passwords are hashed using bcrypt for security.

The high-level architecture flows like this: users interact with the React frontend, which communicates with the Express API, which in turn manages data in MongoDB while storing images in AWS S3. It’s a clean, logical flow that keeps each component focused on what it does best, like a well-orchestrated pit crew where everyone knows their job and executes it flawlessly.

IV. DATABASE DESIGN

The core collections in MongoDB are:

- Users: Stores user profiles, authentication data, and preferences.
- Animals: Each document represents a pet or wild animal sighting, with fields for species, description, location (geo-coordinates), time, and photos.
- Spots: Records individual animal sightings, linked to users and animals.
- (Deprecated) Areas: Originally intended to define neighborhoods, but later replaced by dynamic map-based areas.

Our database schema centers around several core collections that represent the essential elements of our platform. The Users collection stores profiles, authentication data, and user preferences—the basic information that makes each account unique and secure. The Animals collection is where things get interesting, with each document representing either a pet or a wild animal sighting, including species information, descriptions, precise location coordinates, timestamps, and associated photos.

The Sightings collection records individual animal observations, creating connections between users and animals while

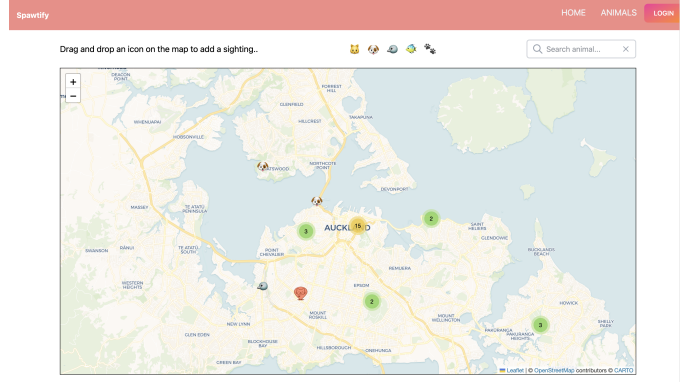


Fig. 4. A screenshot of our home page, showing the interactive map component

building a historical record of where and when animals have been spotted. Comments allow users to engage in discussions about sightings, share additional information, and build the sort of community knowledge base that makes platforms truly valuable.

We also originally planned an Areas collection that would define neighborhoods and animal roaming territories, but this turned out to be one of those ideas that sounds brilliant in theory but proves to be a nightmare in practice. Pre-defining and hard-coding geographical areas would have made the platform rigid and difficult to maintain, like trying to run a modern city with a medieval road system. Instead, we abandoned this approach in favor of letting users define roaming areas dynamically using the Leaflet map interface, which proved far more flexible and user-friendly.

V. USER INTERFACE

The user interface design process began in Figma, where we crafted a visual language that would be clean, friendly, and inviting without being saccharine or childish. We settled on a color palette dominated by white and pink, creating a modern aesthetic that feels approachable rather than intimidating. The interface needed to be fully responsive because people don’t just use computers anymore—they use phones, tablets, laptops, and probably their smart refrigerators if we’re being honest about where technology is heading. The Homepage centers around an interactive map that displays recent animal sightings as emojis, giving users an immediate visual overview of neighborhood animal activity. Click on any marker and you get a popup with details about that particular sighting, creating the sort of immediate gratification that keeps people engaged. The SpotPage allows users to report new sightings with a straightforward form that captures location, time, species, and photos without leaving the page.

The AnimalPage provides a detailed view of specific animals, including their documented roaming areas, historical sightings, and user comments. This creates a sort of biography for each animal, building a picture of their habits, territory, and interactions with the community. The EditAnimalPage allows users to update animal information and adjust roaming areas

directly on the map, providing the sort of real-time editing capability that makes the platform feel alive and responsive.

We chose a map-centric design because location is absolutely fundamental to animal sightings and lost pet reports. You can't find a lost cat by knowing it's "somewhere in the city," but you can hopefully trace it by knowing it was spotted three blocks north of Albert Park at 2:30 pm. The combination of React and Leaflet creates a highly interactive, real-time experience that updates as new information comes in.

VI. IMPLEMENTATION CHALLENGES

Every project encounters problems that make you question your life choices, and Spawtify was no exception. Our first major challenge involved the abandoned Areas collection, which seemed like such a good idea until we tried to implement it. The concept of pre-defining geographical areas for animals and users sounds logical, but in practice it created a rigid system that couldn't adapt to the organic way that animals actually move through urban environments. The solution was to abandon the Areas collection entirely and instead use the Leaflet map to allow users to define and display animal roaming areas dynamically. This approach proved far more flexible and intuitive, allowing users to draw boundaries that actually reflect how animals behave rather than how we think they should behave.

Our second major challenge involved converting user-entered addresses into map coordinates, a process that sounds simple but involves the sort of technical complexity that makes you appreciate why geocoding services exist. Users don't think in terms of latitude and longitude coordinates; they think in terms of "near the corner shop" or "behind the school." We integrated a geocoding API ('react-geolocated') that translates physical addresses into coordinates, allowing the map to display sightings accurately without requiring users to become amateur surveyors.

The third challenge centered on file uploads, which initially seemed straightforward but proved to be one of those technical rabbit holes that consume entire weekends. We started with multer, which I learnt from COMPSCI 719, for handling file uploads in the backend, which worked perfectly in development but proved unsuitable for deployment on servers due to storage cost. The solution involved switching to AWS S3 (another tech I learnt from INFOSYS 735 last semester), which provides an affordable, durable and scalable system for handling user-uploaded images. The frontend now uploads images directly to S3 using pre-signed URLs, while the backend stores the image URLs in MongoDB. This approach keeps the application fast and secure while handling the sort of traffic spikes that occur when many users post a particularly adorable cat photo at the same time.

VII. DEVELOPMENT METHODOLOGY

Managing a project of this complexity required a development methodology that could balance structure with flexibility. We adopted a Kanban-inspired workflow using GitHub Projects, which provided visual organization without the rigid

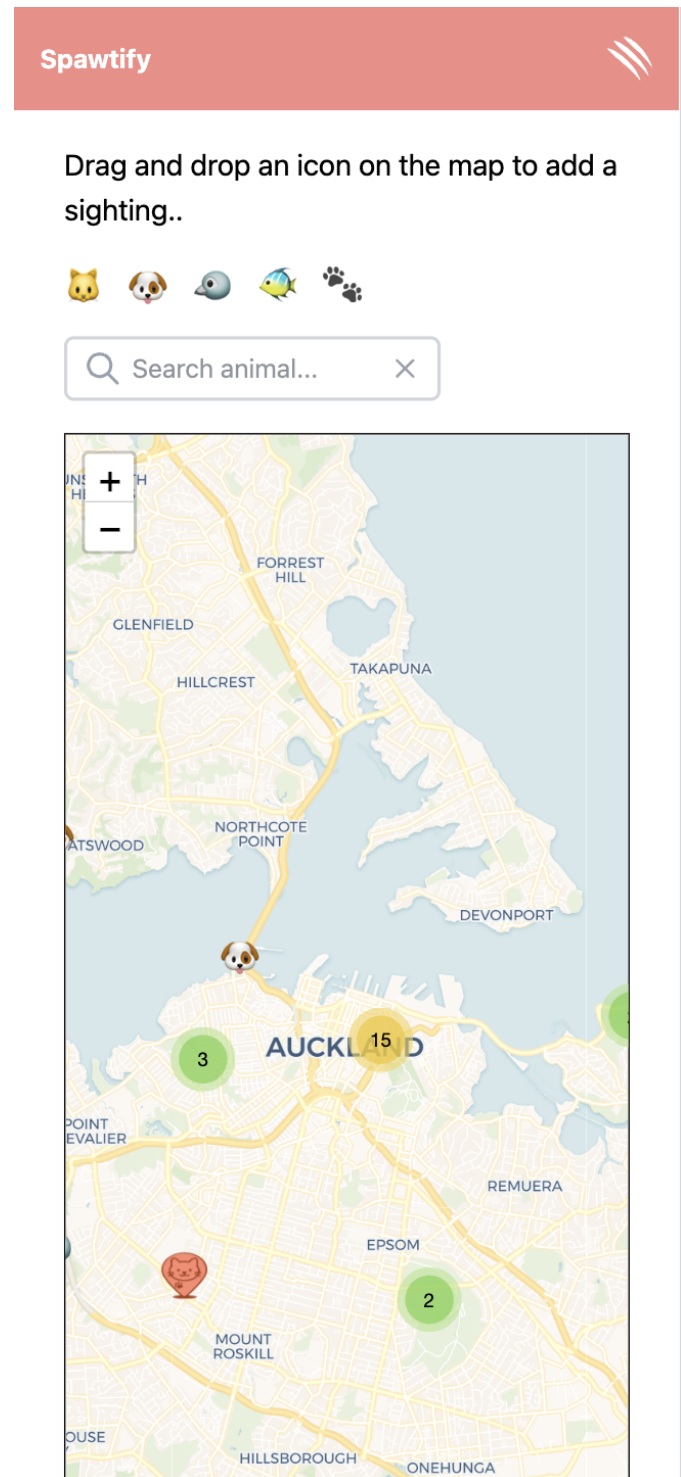


Fig. 5. Responsive design of the frontend, showing how it looks on a mobile device

ceremony of more formal project management approaches. GitHub Projects allowed us to track issues, organize tasks into logical stages, and maintain visibility into who was working on what without requiring daily status meetings that make everyone want to hide under their desks.

We implemented a code review process that required at least two approvals for pull requests, ensuring that changes were reviewed by multiple team members before being merged. This approach caught bugs early, shared knowledge across the team, and maintained code quality without slowing development to a crawl. We used Discord for daily communication, quick questions, and sharing progress updates, creating an informal but effective communication channel that kept everyone connected.

VIII. TECHNOLOGY TOOLS

The development process benefited enormously from modern Gen-AI tools, particularly Cursor IDE and ChatGPT, which transformed certain aspects of coding from manual work into more efficient and enjoyable processes. Cursor's advanced agent mode proved especially valuable for repetitive tasks like generating dummy data, creating test scripts, and building backend APIs. The AI assistance also made frontend CSS work significantly easier, helping to translate visual design concepts into functional code without getting bogged down in the minutiae of flexbox properties and grid layouts.

Traditional resources like React documentation, Stack Overflow, and Reddit filled in the gaps where AI tools fell short, providing real-world solutions to problems that had been solved by real developers facing similar challenges. This combination of Gen-AI assistance and good-old community resources created a development environment that was both efficient and educational.

IX. TESTING

Reliability was crucial for a platform where people would be entrusting us with information about their beloved pets, so we implemented comprehensive testing using Vitest in the backend. Our test suite covered core functionality including user authentication and authorization, CRUD operations for animals and sightings, image upload and S3 integration, and map data handling with geo-coordinates.

Tests were organized by feature rather than by file structure, making it easier to understand what functionality was being tested and why. We used mock data and API calls to ensure coverage without depending on external services during testing, creating a reliable testing environment that could run quickly and consistently.

Frontend testing relied more heavily on manual testing, focusing on user flows and responsive design across different devices and screen sizes. While automated frontend testing would have been ideal, the interactive nature of the map interface and the complexity of user workflows made manual testing more practical within our development timeline.

X. RESULTS AND IMPACT

Spawtify successfully delivers on its core objectives by providing a user-friendly platform for reporting and browsing animal sightings, supporting lost pet recovery through real-time updates and community engagement, and encouraging neighborhood participation in urban ecology. The platform creates a space where pet owners can visualize their animals' roaming areas, community members can contribute to a shared understanding of local wildlife, and neighbors can help each other when pets go missing. The map-based interface makes complex geographical data accessible to anyone who can use a smartphone, while the community-driven approach harnesses the collective knowledge and observation power of entire neighborhoods. Users can see patterns in animal behavior, track seasonal changes in wildlife activity, and contribute to a growing database of urban ecological information.

Perhaps most importantly, Spawtify demonstrates how technology can strengthen community connections rather than replace them. By giving people a reason to notice and care about the animals around them, the platform creates opportunities for neighbors to interact, share stories, and work together toward common goals.

XI. LESSONS LEARNED

Building Spawtify taught us several important lessons about both technical implementation and community platform design. The importance of flexibility in database design became clear when we had to abandon the rigid Areas collection in favor of dynamic, user-defined map component. This experience reinforced the principle that data structures should reflect how users actually behave rather than how we think they should behave.

We learned that user feedback is invaluable for identifying problems that aren't apparent from the development side. Features that seem obvious and intuitive to developers can be confusing or cumbersome for actual users, making early and frequent user testing essential for creating truly usable interfaces.

The power of community-driven platforms became evident as we watched users contribute content, share information, and help each other in ways we hadn't necessarily anticipated. Successful community platforms create value that emerges from user interactions rather than being imposed by platform design, suggesting that flexibility and responsiveness to user needs are more important than rigid feature specifications.

XII. FUTURE DIRECTIONS

Looking ahead, several exciting opportunities could transform Spawtify from a promising prototype into a truly impactful community platform. AI-assisted species identification could help users identify animals from photos, making the platform more accessible to people who aren't naturally gifted at distinguishing between different types of birds or recognizing cat breeds.

Direct messaging capabilities would enable users to communicate about sightings and lost pets without having to

share personal contact information publicly. A missing pet alert system could flag animals as lost and send targeted notifications to nearby users, creating a neighborhood-wide early warning system for pet emergencies.

Administrative tools for content moderation and user management would ensure the platform remains safe and useful as it grows. A native mobile app built with React Native would provide even greater accessibility, allowing users to report sightings and check for updates while they're actually out in their neighborhoods observing animals.

The missing pet alert system represents perhaps the most significant opportunity for community impact. By allowing pet owners to flag animals as missing and automatically notifying nearby users, Spawtify could become an essential tool for pet recovery that harnesses the power of community observation in real-time.

XIII. CONCLUSION

Spawtify represents more than just a technical achievement or a successful class project. It demonstrates how thoughtful technology design can address real community needs while strengthening social connections. By focusing on inclusivity, ease of use, and genuine utility, we created a platform that serves everyone from dedicated pet owners to casual animal observers.

The project taught us to balance technical ambition with user needs, to embrace change as we learned from real-world use, and to recognize that the best community platforms grow organically from user interactions rather than being imposed from above. Most importantly, it showed us that animals can serve as a powerful medium for bringing people together, creating shared experiences and common ground in increasingly fragmented urban environments.

Spawtify proves that technology can enhance rather than replace human community, that complex problems can be addressed through simple, intuitive solutions, and that sometimes the best ideas come from watching a ginger cat disappear over a fence on a cold Auckland morning. In the end, that's probably the most important lesson of all: the best technology solutions often emerge from the most human moments of curiosity and connection.

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